Abdelrahman **Shaapan**

PRODUCT & INDUSTRIAL DESIGNER · LEAD DESIGNER, ELARABY GROUP

I take products from the first sketch and a hand-cut mock-up to the finished object on the shelf.

15+ products in market with ELARABY Group (Tornado, Heller)

20+ features on international design media: Yanko Design, Trend Hunter, BGR and more

4 freelance pieces built and installed, Egypt and Saudi Arabia

Think with my hands

Renders come last. Most of the decisions in these projects were made on paper, in foam and in cardboard, and checked with a real hand, a real wall or a real car before any final CAD.

01

Research

Who uses it, where, and what is already on the market. Heritage, competitors, the standards it has to meet.

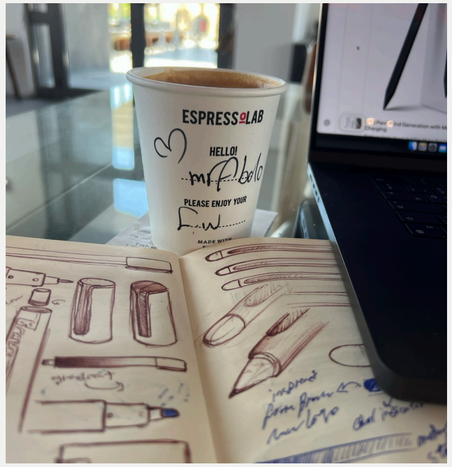
Ramses archive · IMOTEC switch audit

02

Sketch

Pen and paper first, then iPad. Many small ideas fast, then the few worth building.

Bravo sketchbooks · Gamila · The Arrival



03

Make it by hand

Foam, cardboard and printed models to feel size, grip and weight early, when changes are cheap.

Remote foam mock-up · EV holder · UFO model

04

Test

Prototypes in the real context: in the hand, on the wall box, on the car glass, against the calipers.

Silicone keys · IMOTEC wall mock-up · IQ Ring QC

05

Make it real

Production CAD, print-ready parts, CNC cut files and drawings, then follow-up with the factory.

Midmar by FATCO · W-01 · 119 + 39 CNC parts

Selected work

01	Tornado Smart TV Remote	ELARABY GROUP	IN MARKET	2025
02	Ramses Gamila	PHOTONX · CAIRO DESIGN WEEK	COMPETITION	2026
03	Midmar	ALULA CAMEL RACE · FATCO	BUILT	2024
04	The Arrival	ARTIST ALI KAMEL	BUILT	2024
05	Bravo Edge	BRAVO · FREELANCE	CLIENT CONCEPT	2025
06	W-01	VERTEX STUDIO	BUILT	2026
07	IMOTEC Smart Switch	PHOTONX · IMOTEC	PROTOTYPED	2026
08	EV Charger Holder	PRODUCT LAB, COVENTRY UNIV.	PROTOTYPED	2026
09	IQ Ring	FREELANCE CLIENT	BUILT	2024–25
10	Key concepts	PUBLISHED ON BEHANCE	CONCEPT	2020–24

01 · IN MARKET · 2025

Smart TV **Remote**

One universal remote for every new Tornado TV, designed from scratch and found by touch.

THE BRIEF

A single remote that works across all upcoming Tornado TV models: simple, intuitive, and in line with Google's TV interface requirements.

ROLE

Industrial design, ergonomics, prototyping

COMPANY

ELARABY Group · Tornado

PROCESS

Paper sketch, foam, silicone prototype

STATUS

In market, 2025

Shaped in the hand

I wanted the remote to be used without looking at it, so every step was tested by holding it, not on screen.

01 Hand-carved foam

From the paper sketch straight to foam, to set length, thickness and where the thumb rests before any CAD.

02 Silicone 3D-printed prototype

Printed in silicone to test how the buttons feel and press, not only how they look.

03 Found by touch

Home and Search are concave so the thumb finds them blind. The rest are convex, so they are easy to learn.

04 Braille for everyone

Braille marks sit next to the essential keys: for visually impaired users, and for anyone in a dark room.

THE PRODUCTION REMOTE, IN MARKET 2025

CONCAVE HOME AND SEARCH KEYS

"Though small in size, it took a long and thoughtful process to develop."

One remote for the whole Tornado TV range. It ships with the brand's smart TVs.

Made, sold, taken home

A product only succeeds when it is made, people choose it, and they take it home. As Lead Designer at ELARABY Group, more than fifteen of mine have, under Tornado, Heller and Sharp. A few of them:

Gas Water Heater

Tornado · 2023.

Cooker

Tornado · 2024. One of the first Tornado products exported to the Gulf.

Twin-Tub

Tornado · 2025. An affordable washer for every Egyptian home.

Twin-Tub II

Tornado · 2025. A stronger brand language that stays easy to make.

Smart TV Remote

Tornado · 2025. One remote for the whole TV range.

Freestanding Cooker

Heller · 2025. For Germany, launched at IFA Berlin.

Hob & Hood

Heller · 2026. A built-in gas hob and a chimney hood.

02 · COMPETITION · CAIRO DESIGN WEEK 2026

Ramses Gamila

What if Ramses never stopped? An electric city car for an Egyptian family of four.

THE QUESTION

Ramses was Egypt's own car, built at the foot of the Giza pyramids from 1960. What would it look like today if production had never stopped: an evolution, not a retro copy?

MY ROLE

Exterior design

TEAM

Youssef El Araby (concept, creative direction) · Lina Abdelraouf (interior)

SIZE

4700 × 1780 × 1445 mm

BATTERY

36 kWh, 60 fixed / 40 swappable

Start from the archive

Before drawing, I collected the original Ramses and Gamila: press, photos and the 1960s wordmark, to find what should carry forward and what should not.

رمسيس

أول سيارة عربية

إن إنشاء مصنع للسيارات فى بلد لا تزال الصناعة فيه ناشئة ليس بالأمر الهين . لذلك واجه السيد جورج حاوى وهو مهندس مصرى إختصاصى ، صعوبات لا حصر لها عندما شرع فى ١٩٥٤ فى وضع مشروع لإنتاج سيارة شعبية صغيرة .

ولم تكن الفكرة هى إنتاج سيارة من أى نوع وبأى ثمن . وقد كانت الموارد الفنية والصناعية محدودة ، وكذلك الأموال اللازمة . فمن المعروف أن إنتاج السيارات بكميات محدودة عمل لا يثير إلا المنتجين الذين تتوفر لديهم أموال طائلة ومعدات كاملة .

لذلك كان من الضرورى أن تصمم السيارة على نحو يناسب الاستعمال الخاص الذى

« سيارة رمسيس » من طراز ليموزين

01 **"The first Arab car"**

Ramses in the press. The car was a national story, so the new one had to feel familiar to Egyptians.

02 **The original Gamila**

Its friendly face and small, upright body are what I kept. The chrome and fins stayed in the 1960s.

03 **Small and upright**

The proportion I wanted to keep: a compact car for a family of four.

رمسيس

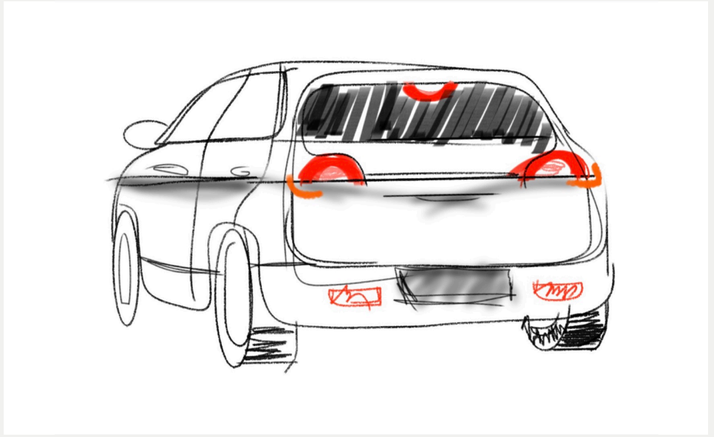
04 **The 1960s wordmark**

Traced from the original source as a link between the old car and the new one.

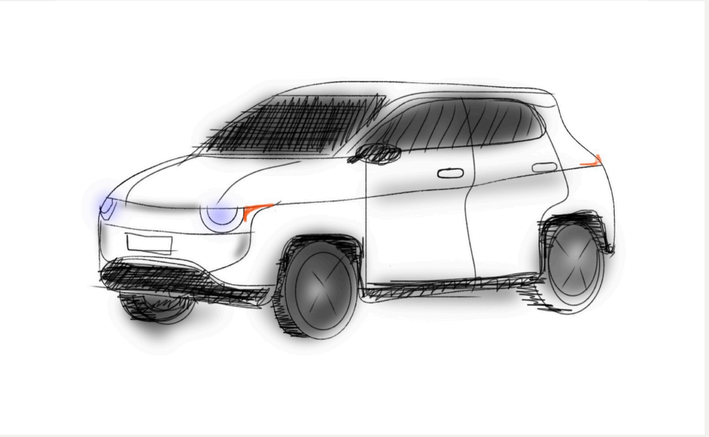
The face is an eye

The idea came on paper: an open lamp with a kohl line that turns into an amber lash, after the painted Eye of Horus, and one line that wraps the whole body.

01 · PEN ON PAPER: THE EYE AND THE SINGLE LINE

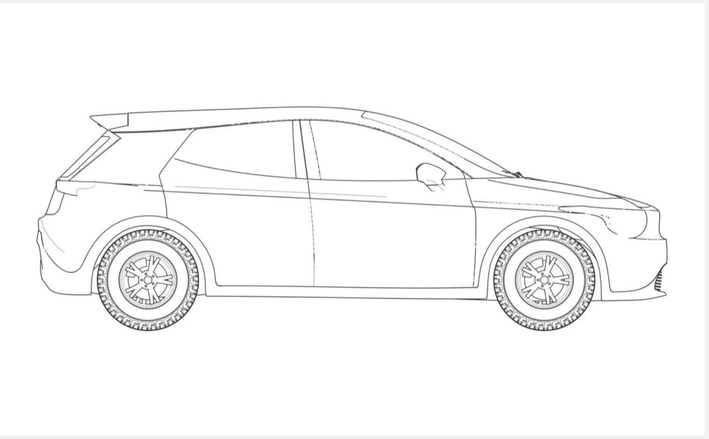


02 · IPAD: REAR AND LAMP STUDIES



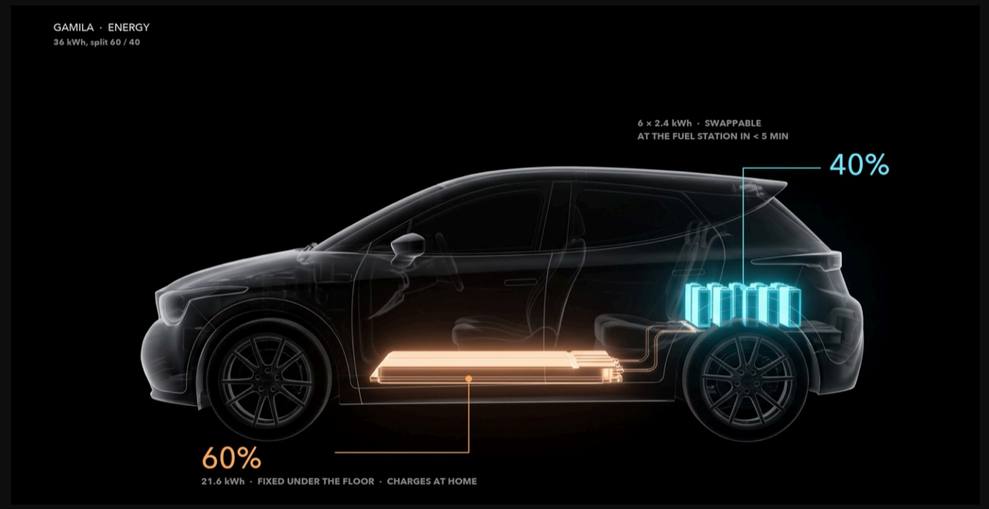
03 · IPAD: THE BODY SIDE

04 · CLAY RENDER OF THE FINAL MODEL



05 · SIDE LINE DRAWING FROM THE 3D MODEL

THE OPEN EYE WITH ITS AMBER LASH



60% UNDER THE FLOOR CHARGES AT HOME · 40% AS SIX PACKS SWAPPED AT A FUEL STATION IN UNDER 5 MIN

PEGBOARD DASH: CLIP-ON ACCESSORIES, SCREEN OPTIONAL, ADDED LATER

03 · BUILT · 2024

Midmar

A screen table for the VIP lounge at the AlUla camel race. Built and installed in April 2024.

THE PROBLEM

Guests sit in a U-shaped majlis and need to follow a race that happens far away. The table has to bring the race to the middle of the room, for every seat.

ROLE

Design, with Mohamed Agag

BUILT BY

FATCO, Riyadh

SIZE

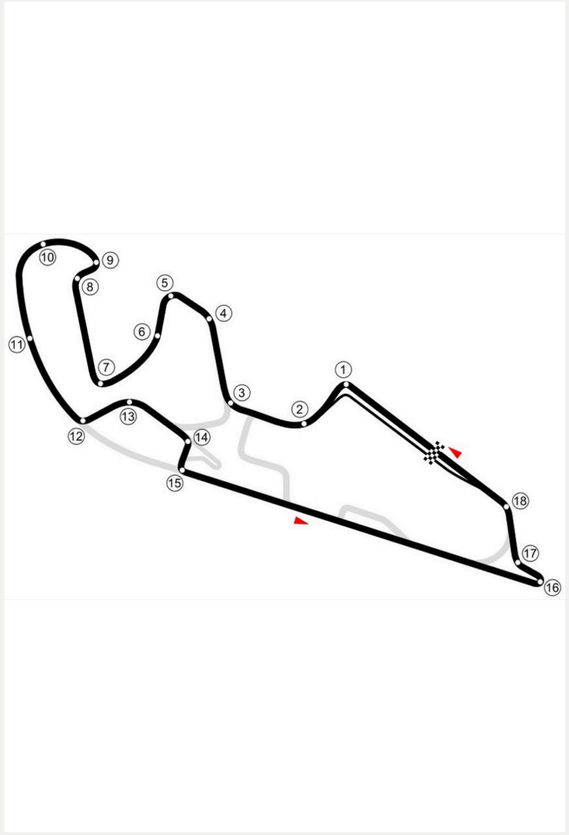
716 × 1091 × 620 mm · 3 screens

FINISH

Espresso veneer, white shell

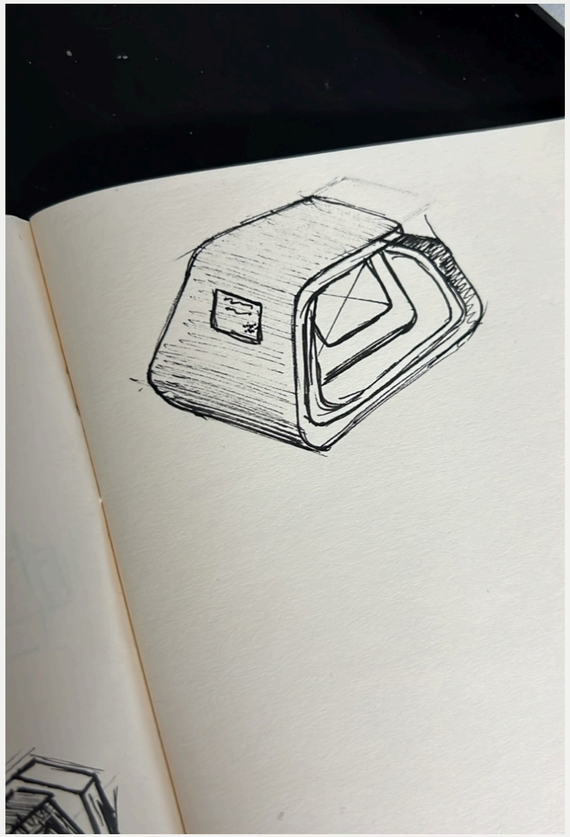
A table drawn like a track

The form started from a race circuit: one continuous band that loops, with a raked profile that carries the speed of the race. Screens face the three sides of the majlis; the centre one is double-sided.



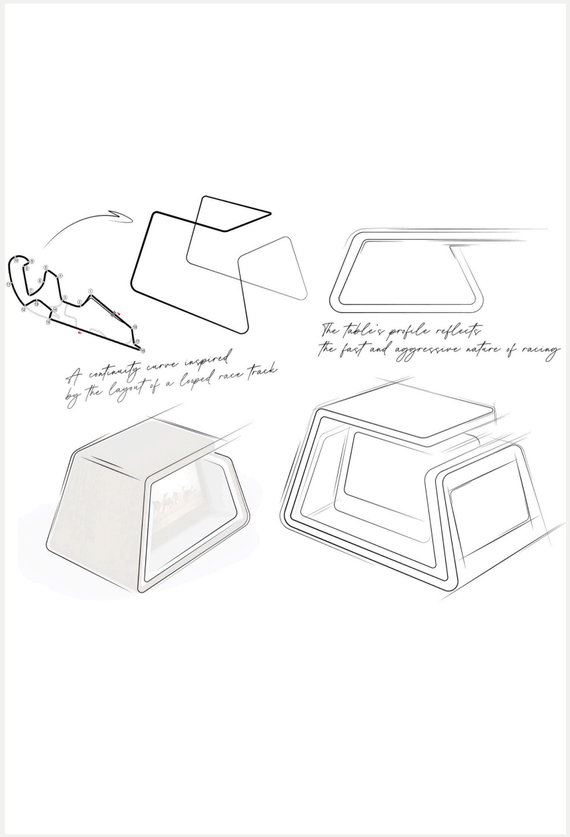
01 **Reference: a circuit**

A closed loop with fast corners became the plan for one continuous band.



02 **Pen sketch**

The loop turned into a trapezoid, raked forward like a car at speed.



03 **Refined sketch**

Band, floating tray and screen opening fixed before CAD.

04 **Shell, tray, screens**

The table broken into a shell, a tray and screen modules for the workshop.

Built by hand at FATCO

The bent shell, the veneer and the raked angle all had to match the design, so the build was checked against it at each stage in the Riyadh workshop.

01 **The shell takes shape**

The looping band built up as one continuous form.

02 **Inner liner**

The white inner shell fitted inside the band.

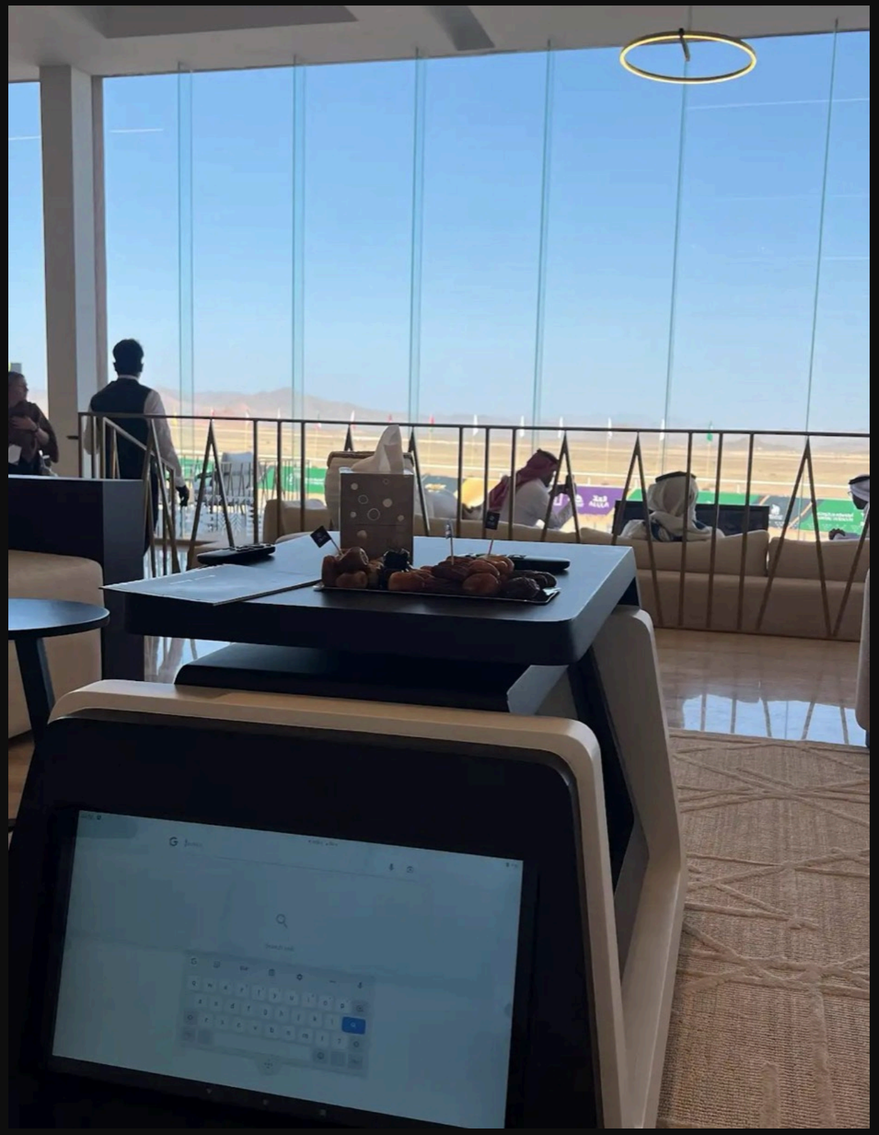
03 **Checking the rake**

The profile angle measured on the real part against the drawing.

04 **Nearly finished**

Dark finish on, tray and screen opening in place.

IN PLACE IN THE VIP LOUNGE, ALULA, APRIL 2024



RACE DAY: THE RACE ON EVERY SIDE OF THE MAJLIS

04 · BUILT · 2024

The **Arrival**

A walk-in spaceship for Ali Kamel's paintings, cut in Egypt and rebuilt in Saudi Arabia for every show.

> THE CONSTRAINT
>
> It had to be made in Egypt, travel to the artist, and go up again at every exhibition. So it had to pack flat and assemble without glue or special tools.

ROLE
Form, parametric slicing, cut files

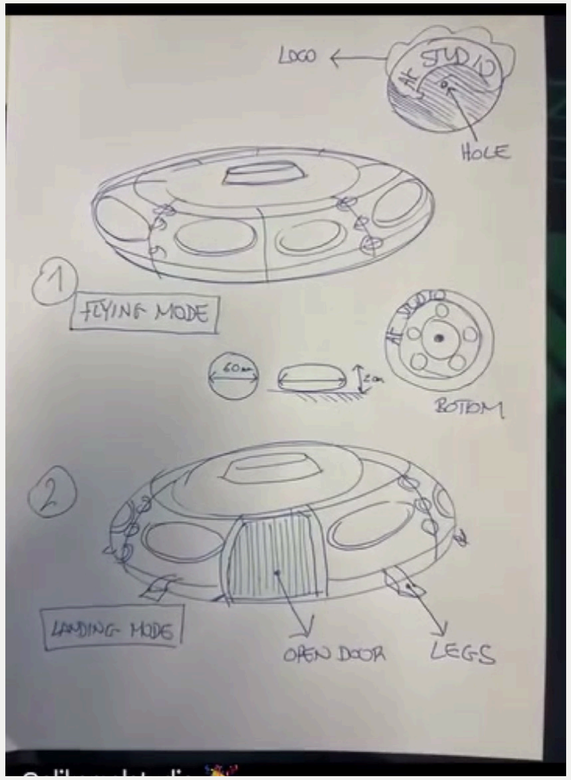
STRUCTURE
119 ribs + 39 rings, slot-fit

TOOLS
Sketch, Rhino, Slicer for Fusion 360

SHOWN
The Arrival, FCR

From a sketch to 158 flat parts

I checked the form at small scale in my hand first, then turned it into a flat-pack structure a CNC can cut.

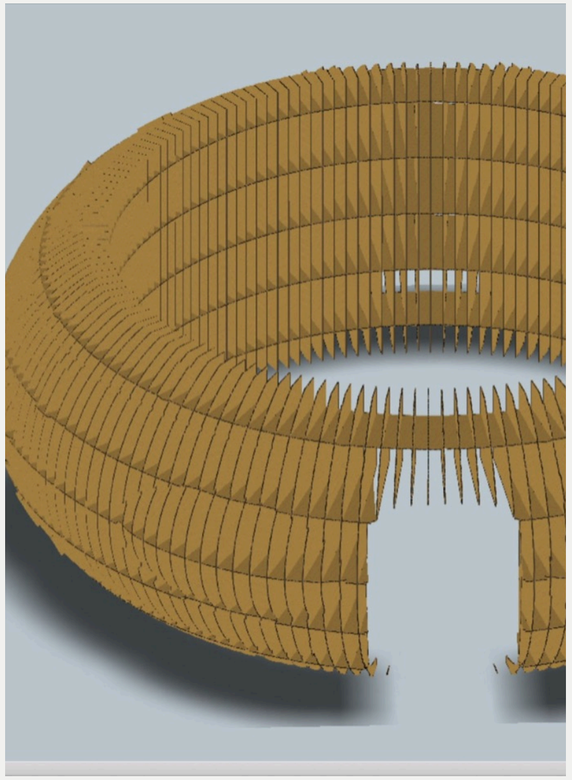


01 **Two modes**

Flying, and landed on legs with an open door. The door had to be a real way in.

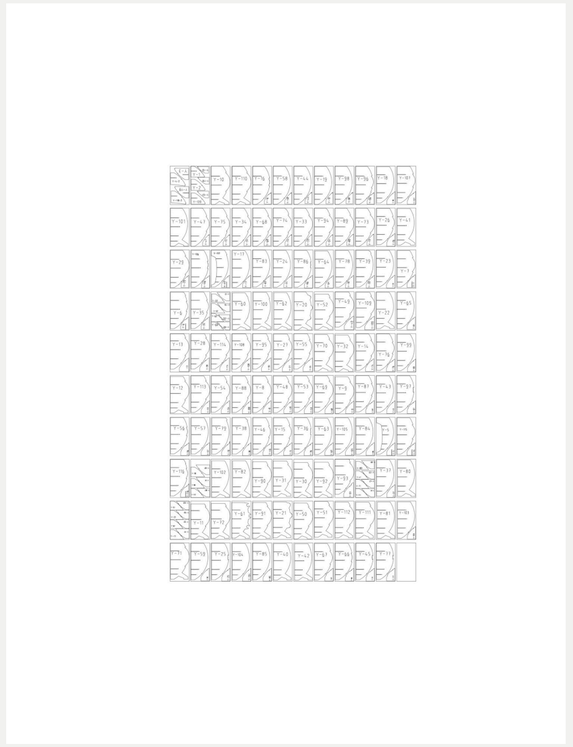
02 **Printed scale model**

A 3D-printed model to judge the proportions with my own eyes, not only on screen.



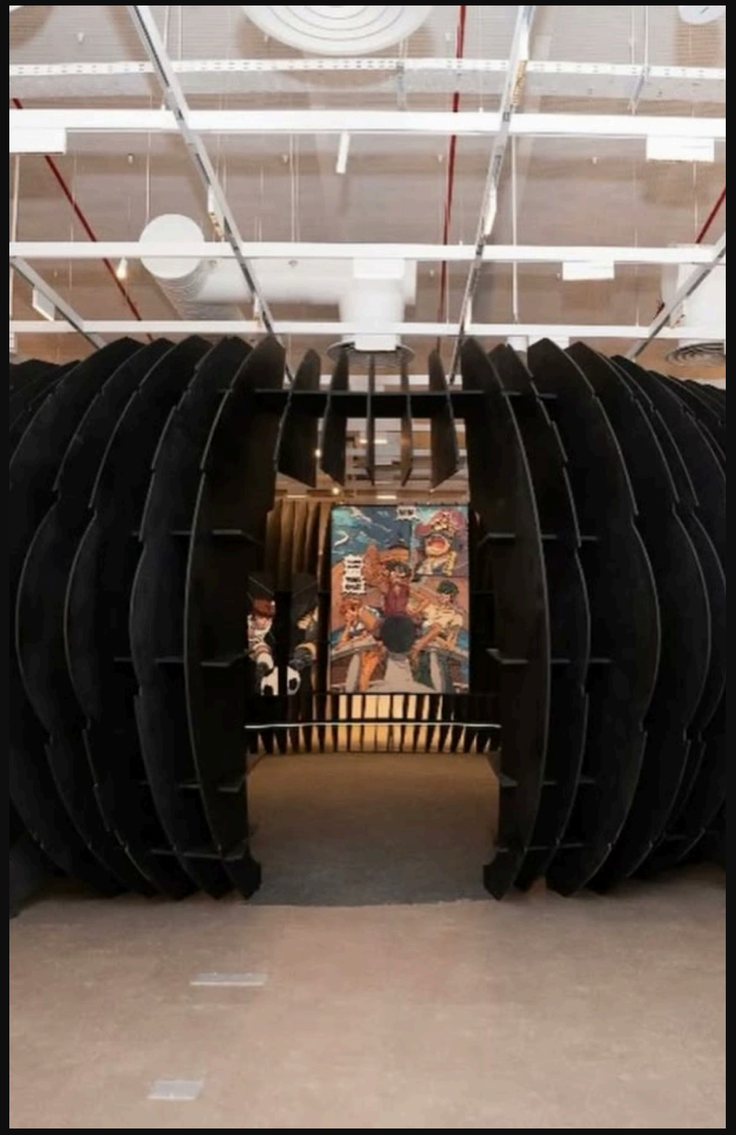
03 **Parametric slicing**

The Rhino form, sliced into 119 radial ribs locked by 39 rings in Slicer for Fusion 360.



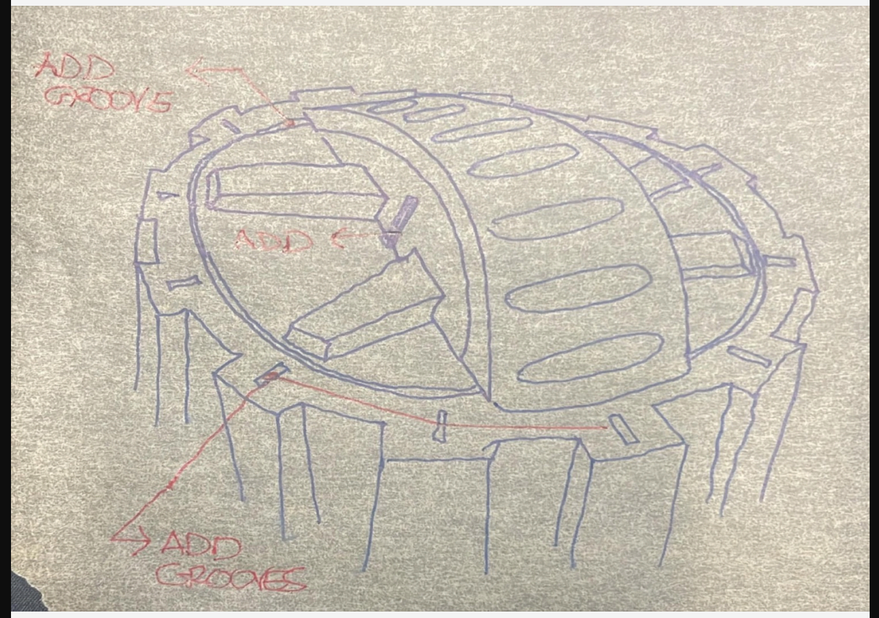
04 **Cut sheets**

Every part nested on flat sheets for the CNC: 13 mm ribs, 17 mm rings, numbered for assembly.



INSTALLED AT FCR

INSIDE, THE RIBS BECOME THE GALLERY WALL



SKETCH: GROOVES WHERE RIBS AND RINGS LOCK

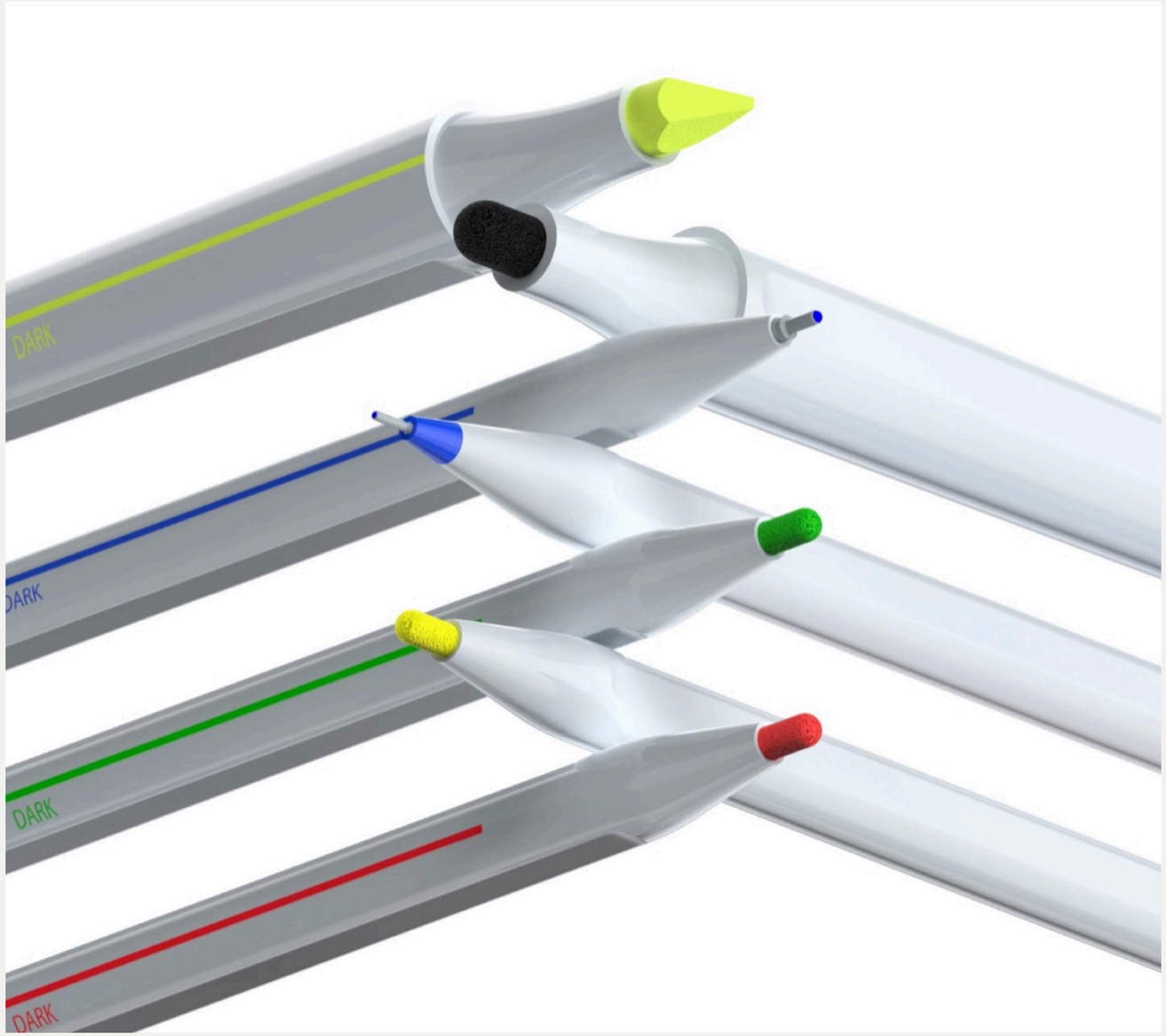
05 · CLIENT CONCEPT · 2025

Bravo **Edge**

A new identity for Bravo's writing instruments: one family, from highlighter to fineliner.

THE BRIEF

A new visual identity across ballpoint and gel pens, colouring pens, markers and whiteboard markers.



ROLE
Product design & identity

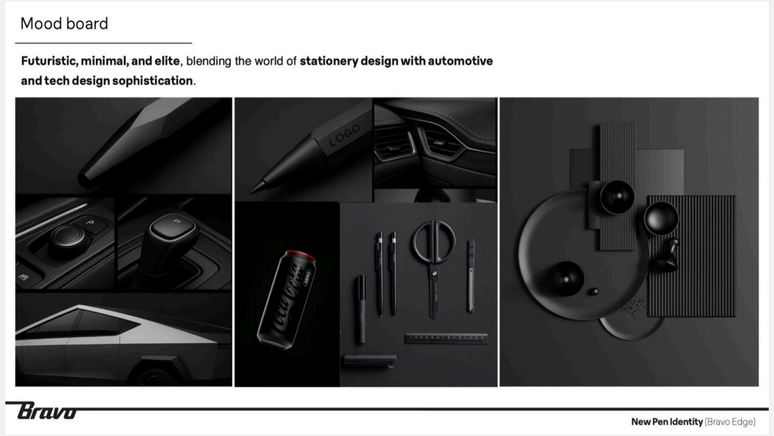
FAMILY
5 instruments, 4 colourways

FINISH
Matte body, gloss logo

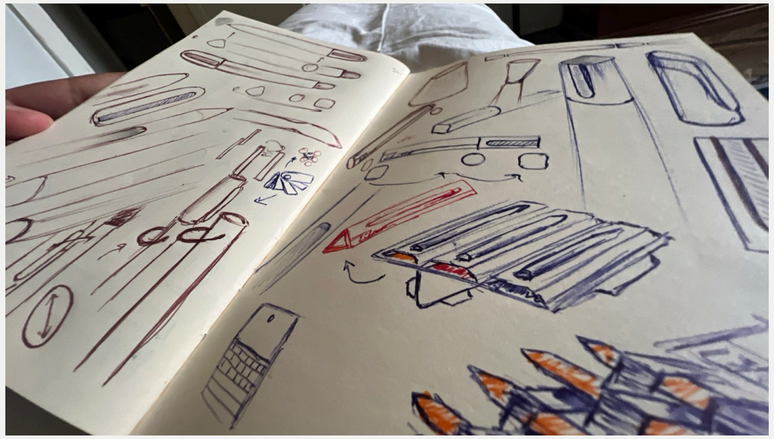
STATUS
Concept 1, presented to client

Pages of pens

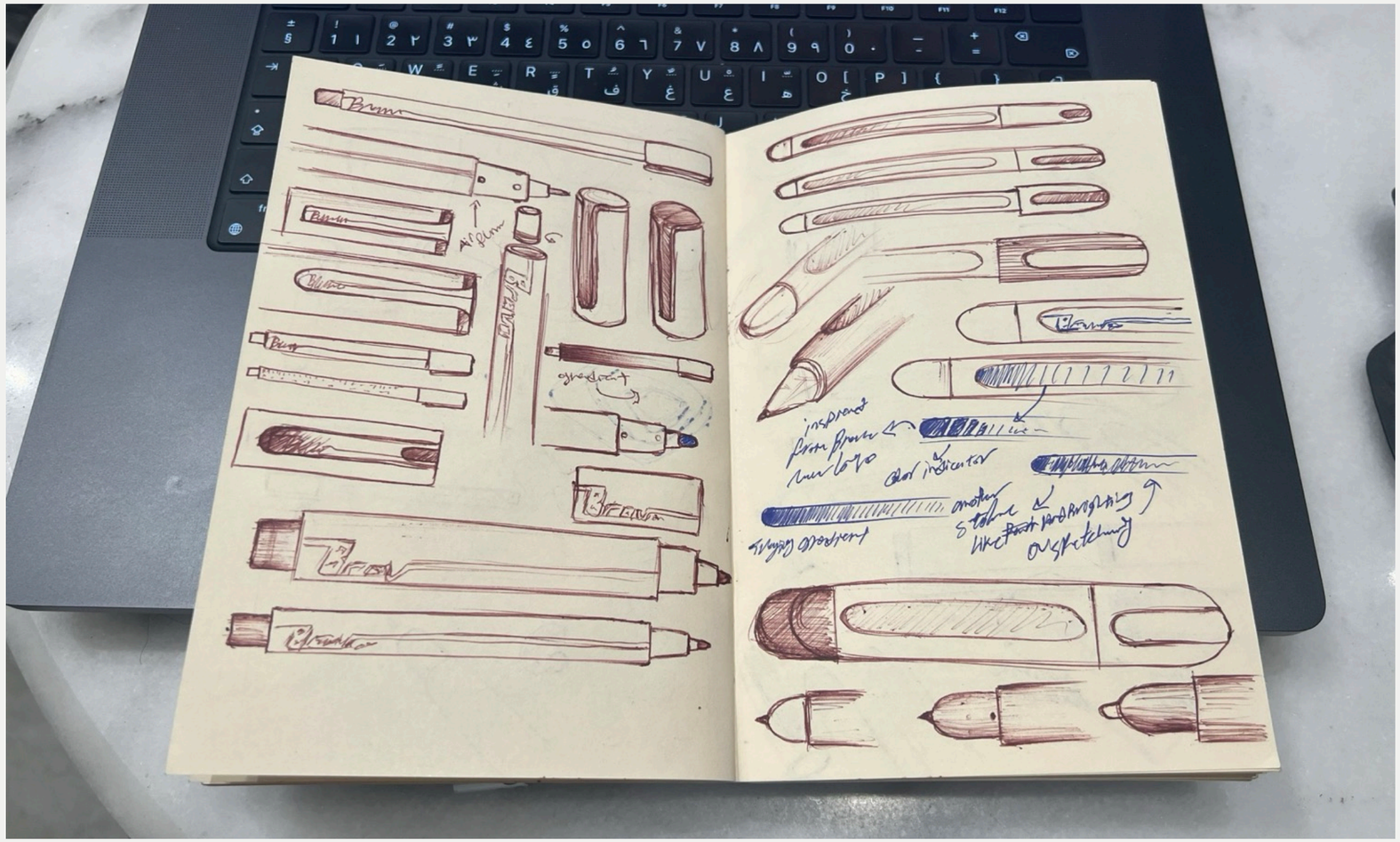
After the brief and a mood board from tech and automotive products, I filled sketchbook pages with caps, clips and sections until one family language appeared.



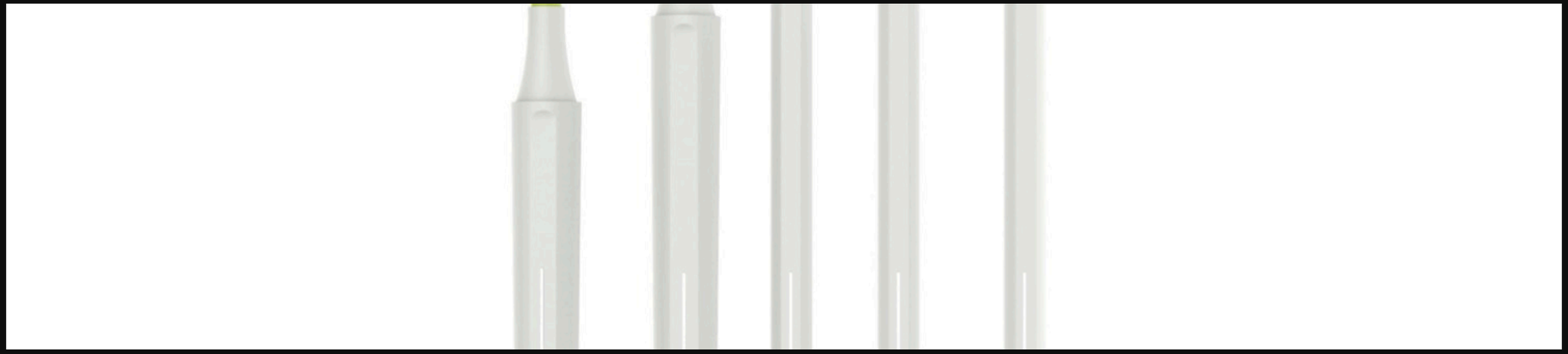
01 · MOOD BOARD: FUTURISTIC, MINIMAL, ELITE



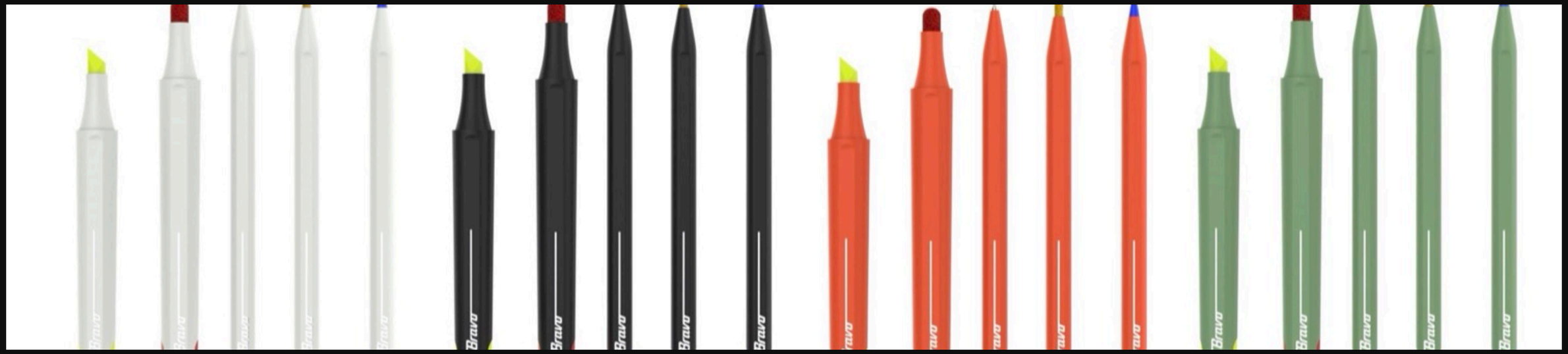
03 · MARKER STUDIES OF THE CHOSEN DIRECTION



02 · SKETCHBOOK SPREAD: CAPS, CLIPS AND SECTIONS



ONE SOFT BODY AND A LONG ENGRAVED LINE ENDING IN THE WORDMARK; THE INK COLOUR SHOWS AT THE TIP AND THE END



THE FULL FAMILY IN FOUR COLOURWAYS

06 · BUILT · 2026

W-01

A 3D-printed desk lamp for Vertex Studio: one base, interchangeable shells.

THE BRIEF

Vertex makes all its home décor on 3D printers. They asked for an identity and the first products that show it as objects.

ROLE

Design direction, product design, print-ready CAD

EDITIONS

Solid, Perforated, Clear, Mini

BASE

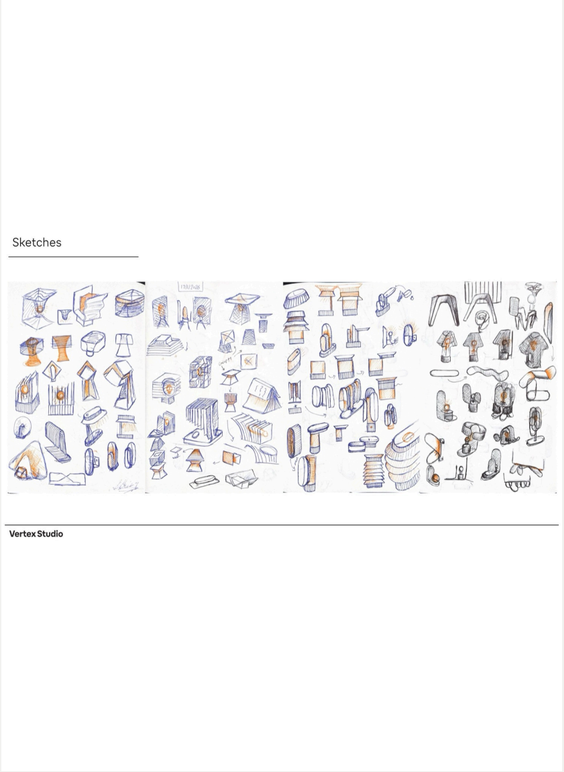
111 × 105 × 60 mm

PROCESS

FDM 3D printing

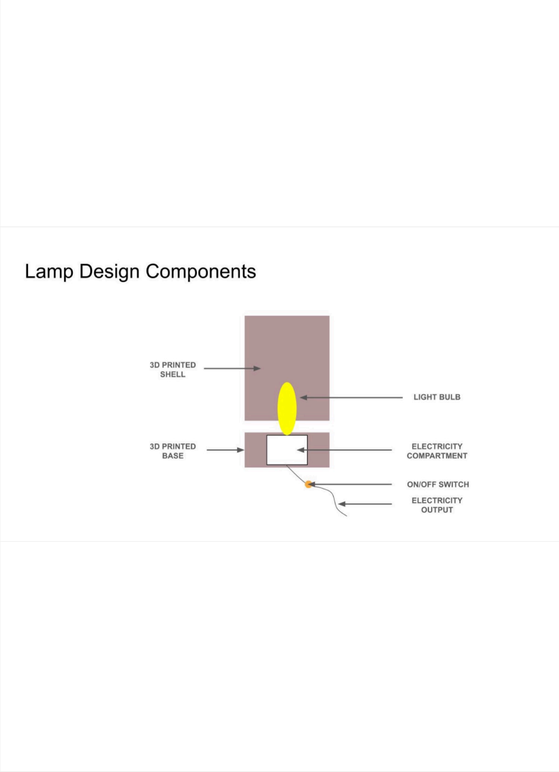
Designed for the printer bed

One idea, a capsule of light on a solid base, then every decision bent to what a home printer can make well.



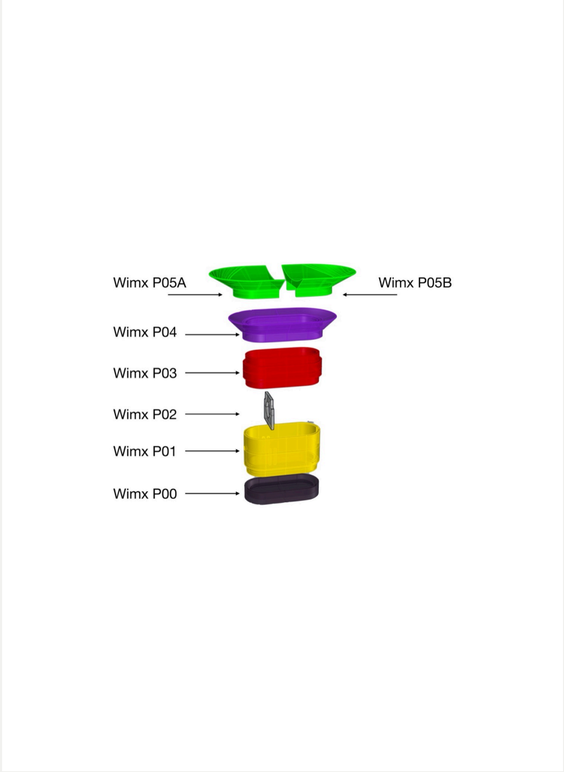
01 Sketches

Rounded, rectilinear and bold: lamp ideas around one capsule shape.



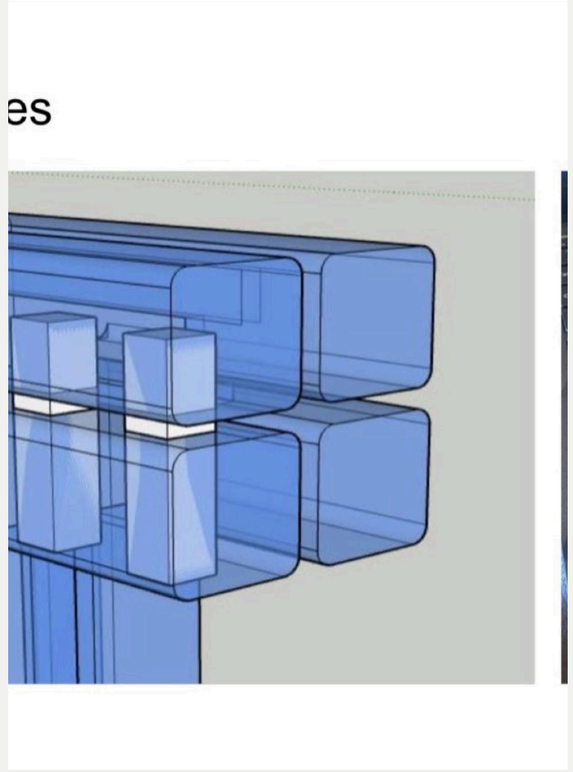
02 What goes where

Bulb, switch and wiring hidden in the base, so the shell can change.



03 Printable parts

The shell split into parts that each fit a home printer bed.

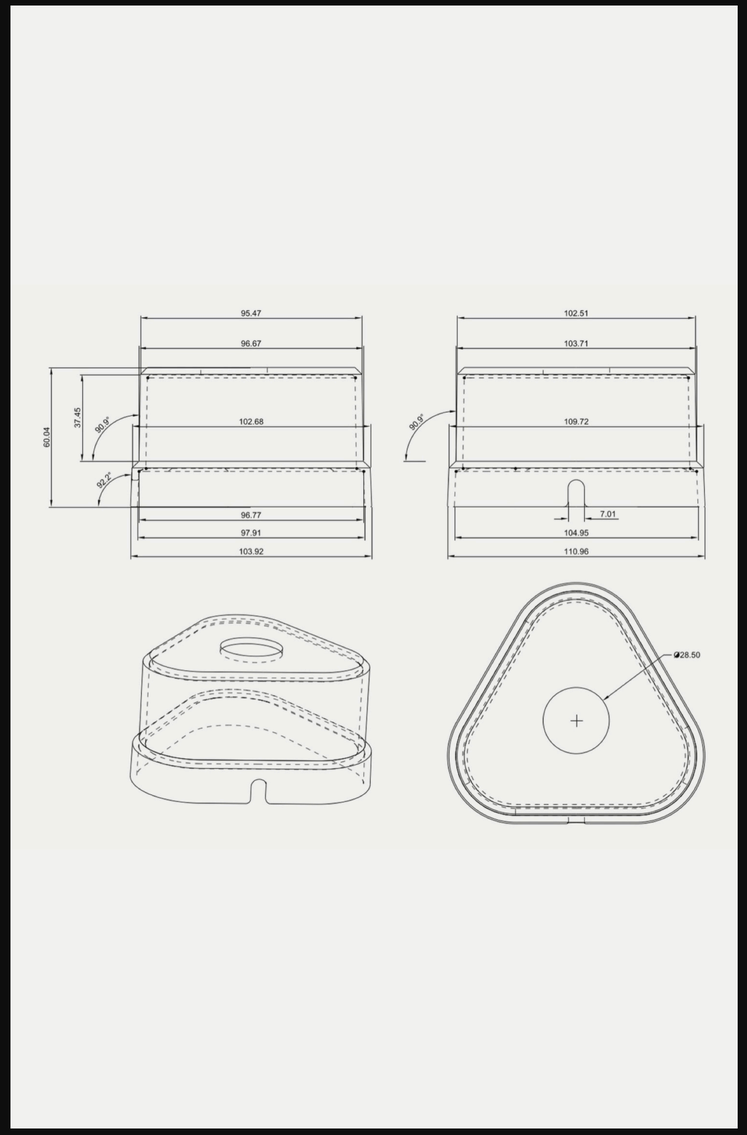


04 Design prototypes

Printed test pieces checked by hand before the final files.

THREE SHELLS, ONE BASE: CLEAR, SOLID, PERFORATED

PERFORATED EDITION THROWS ITS PATTERN



DRAWING: SHARED BASE, Ø28.5 MM LAMP SEAT, CABLE SLOT

07 · PROTOTYPED · 2026

IMOTEC **Switch**

A Wi-Fi smart switch that fits the wall box you already have.

THE BRIEF

A light and shutter switch for any home with no new wiring, from the premium end down to the upper mid-range.

ROLE

Research, concepts, ID, CMF, renders

RANGE

1–4 gang + shutter switch

CONCEPTS

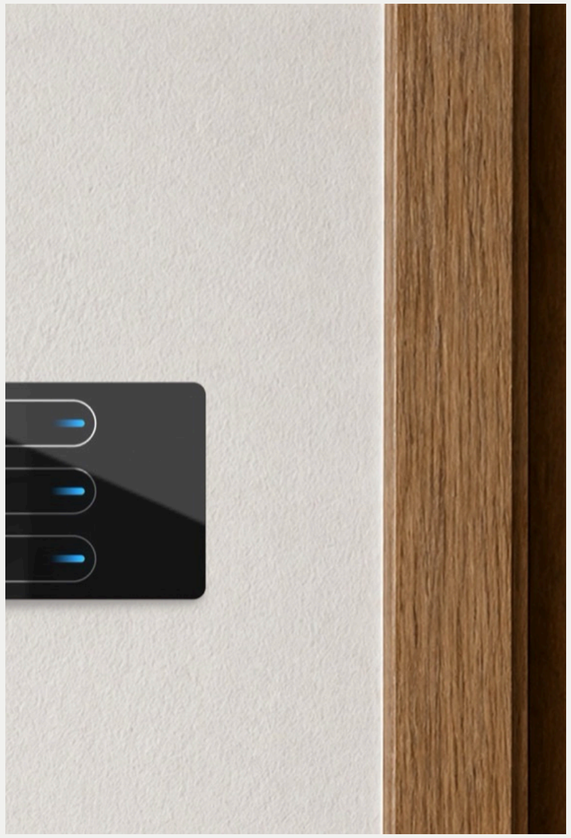
4 directions, 13 variants

STATUS

Prototype on a wall mock-up

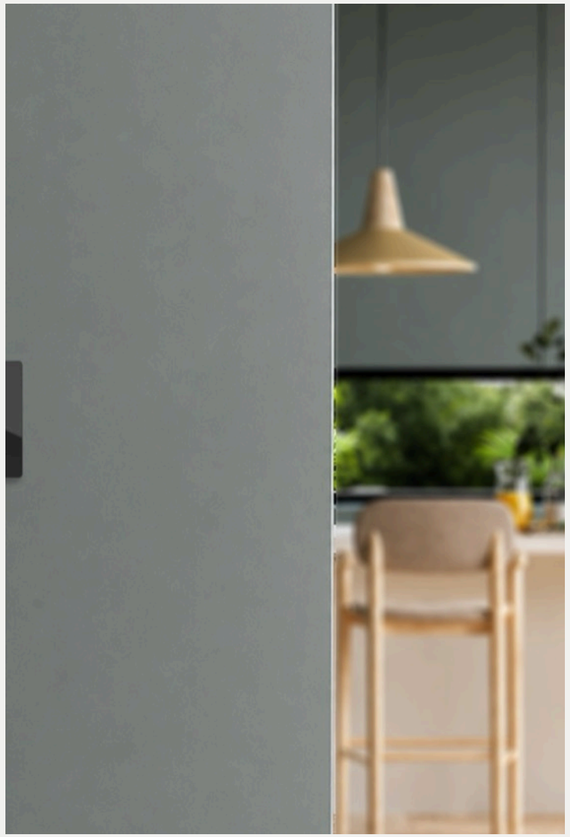
Four ways to press a light

Research covered IMOTEC's own range, the brands already on the wall and how each switching technology feels. It became four concepts, each tried as a 1 to 4 gang set.



01 **Piezo glass**

Concept 1: no moving parts, a press on solid glass.



02 **Light line**

Concept 2: a soft line of light at the key edge, the direction we developed.

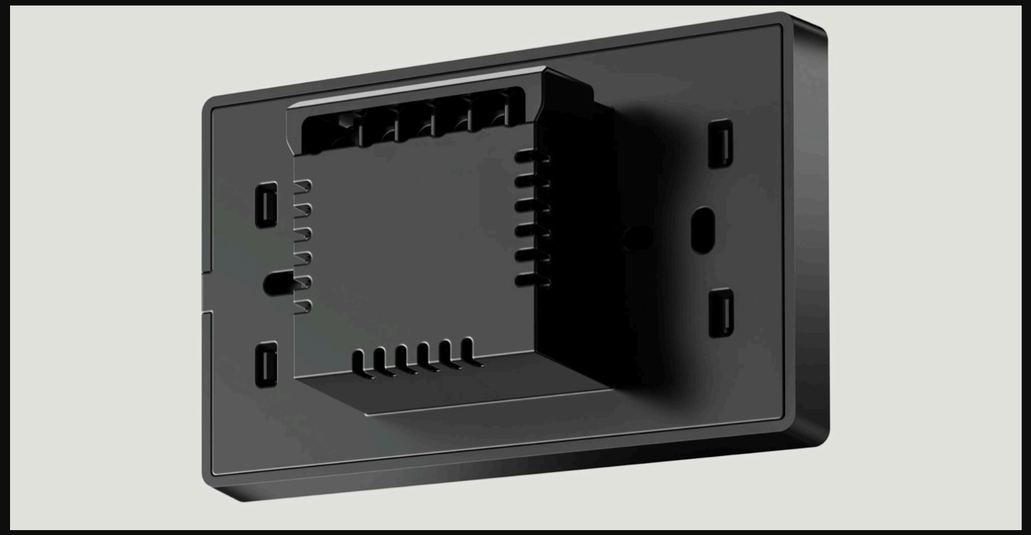
03 **More directions**

Every concept was tried as a 1, 2, 3 and 4 gang set: 13 variants in all.

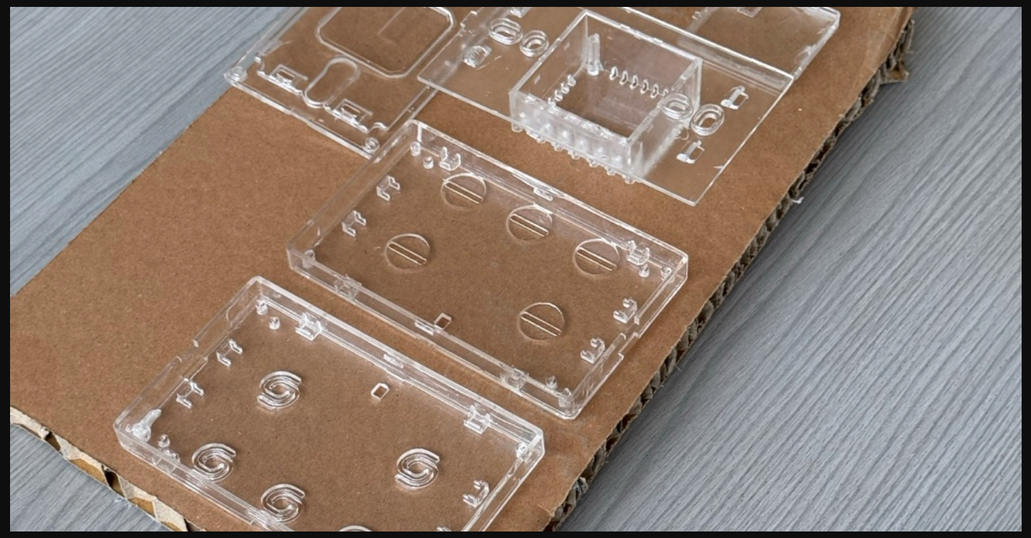
04 **Tested on a wall**

Prototype parts fitted on a wall mock-up to check the box fit and the frame.

FINAL: ONE CALM BLACK FRAME, GLASS KEYS, A SOFT LINE OF LIGHT AT EACH EDGE



BACK HOUSING DESIGNED FOR A STANDARD WALL BOX



CLEAR PROTOTYPE PARTS

08 · PROTOTYPED · 2026

EV Charger **Holder**

Keeps the charging connector off the ground by gripping the car's own glass.

THE PROBLEM

While an electric car charges, the heavy connector and cable end up on the ground: dirty, a trip hazard, and awkward to pick up.

ROLE

Product design, prototyping

TEAM

With Ahmed, Habiba and Mohamed

GRIP

Two flexible rubber suction pads

FITS

Any car, nothing fixed to it

Tested on a real car

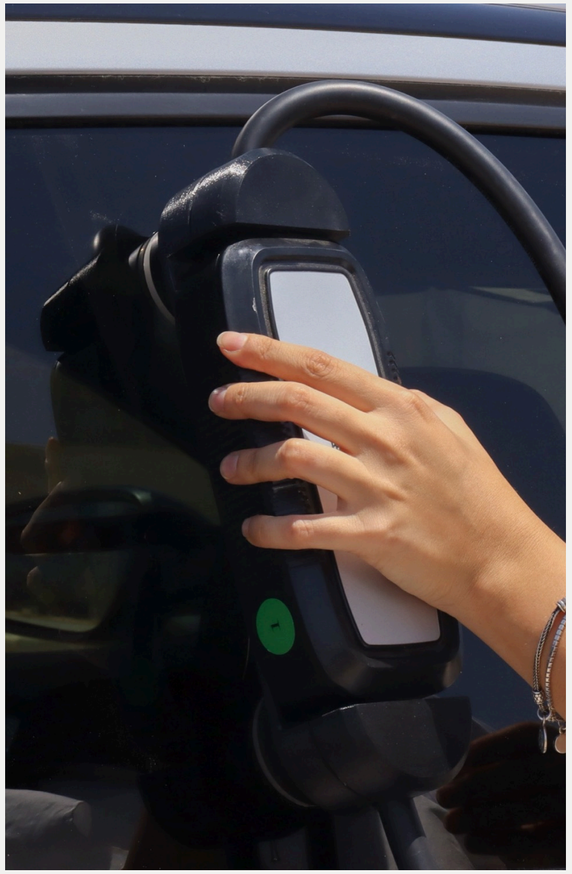
We worked with cardboard first and then a working prototype, and judged every version on an actual EV while it charged.

01 **Cardboard first**

Cardboard pads to find the size and spacing, next to the working prototype.

02 **The suction pads**

Two flexible rubber pads, so the grip follows curved glass.



03 **Mounting**

Pressed onto the window: no tools, and nothing fixed to the car.

04 **In use**

The connector and cable hang clear of the ground while the car charges.

09 · BUILT · 2024–25

IQ Ring

A smart tasbih you wear: a ring with a screen and one button that counts your dhikr.

> THE CHALLENGE
>
> Fit a screen, a battery and a button into a ring, in three sizes, without changing the electronics.

ROLE

Product design, CAD for production

SIZES

18 · 20 · 22 mm, one shared outline

DISPLAY

Count, time, battery

STATUS

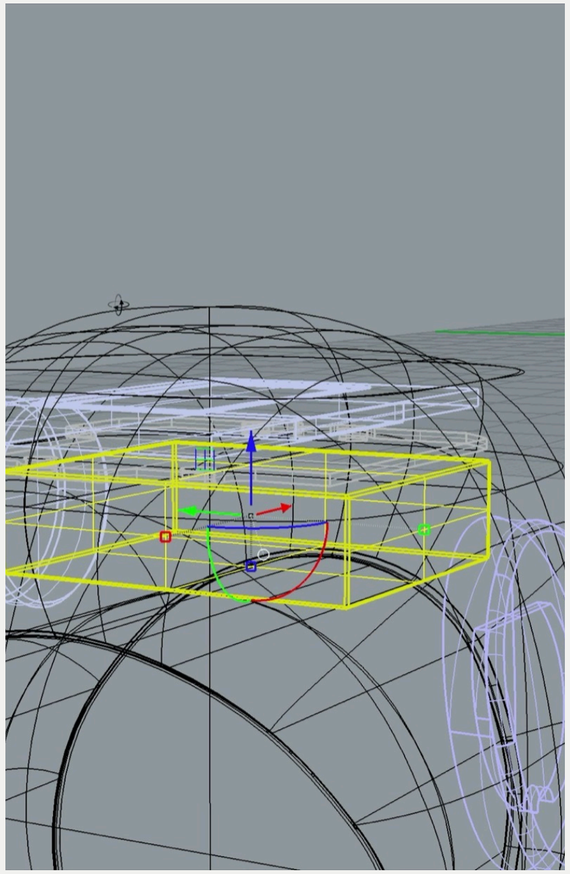
Produced, QC checked

From CAD to the calipers

I took it from early directions to the final Design C, built the production CAD, then checked the factory samples against it.

01 **Explorations**

2024 colour and button studies for the head.



02 **Packaging in Rhino**

Screen and battery placed first, then the head surfaced around the button.

03 **Three sizes**

18, 20 and 22 mm share one outline, so the electronics stay the same.

04 **Production QC**

Samples measured with calipers against the CAD before the run.

Key concepts

Self-initiated work I keep doing alongside client projects. Unofficial concepts, not affiliated with the brands.

PUBLISHED 4 Sep 2024

Game Boy Flip

A small square handheld that unfolds into one wide screen. Rhino, KeyShot.

PUBLISHED 18 Aug 2024

CMF Speaker

A clear-shell portable speaker with a light bar and a player screen. Blender.

PUBLISHED 19 Feb 2021

Flex OLED TV

A panel that flexes from flat to curved on its stand. Maya, KeyShot.

PUBLISHED 15 Apr 2020

BRAUN Audio 07

Braun-style speaker with a touch screen and Spotify built in. 474 appreciations, 13.9K views.

Let's Talk

WEBSITE
abdelrahmanshaapan-07.github.io

EMAIL
abdelrahmanshaapan@gmail.com

BEHANCE
behance.net/Abdelrahman_Shaapan

INSTAGRAM
@abdelrahma